Antibiotics are produced by species of Streptomyces, Bacillus, Penicillium and Cephalosporium.

Infobits

1904 Ehrlich found that the dye try pan red was active against the trypanosome that causes African sleeping sickness and could be used therapeutically.

Drugs such as the sulfonamides are sometimes called antibiotics although they are synthetic chemotherapeutic agents which are not microbially synthesized.

Classification of Antibiotics

The antibiotics are usually classified on the basis of:

- Target group of microorganisms
- Antimicrobial spectrum and
- Mode of action

❖ Classification based on target group of microorganisms

Based on the target group, the antibiotics can be classified as antibacterial, antifungal and antiviral.

❖ Classification based on Antimicrobial spectrum

Antimicrobial spectrum or antibiotic spectrum refers to the range effectiveness of antibiotics on different kind of microorganisms, i.e. the range of different kind of microorganisms that can be inhibited, killed, or lysed by a particular type of antibiotic.

The susceptibility of microorganisms to individual antibiotic varies significantly and on account of this, the antibiotics can be classified in two groups as,

Broad – spectrum antibiotics

These attack different kinds of microbial pathogens and therefore find wider medical use. Antibacterial antibiotics of broad – spectrum are effective against both Gram positive and Gram negative bacteria. They also attack pathogens belonging to *Mycobacteria*, *Rickettsia*, and *Chlamydia*. Similarly, broad – spectrum antifungal antibiotics attack different type of fungal pathogens.

Narrow – spectrum antibiotics

Narrow – spectrum antibiotics are categorized as those that are effective only against a limited variety of microbial pathogens. These antibiotics are quite valuable for the control of microbial pathogens that fail to respond to other antibiotics. For example, vancomycin is a narrow spectrum glycopeptide. It is an effective bactericidal agent for gram – positive penicillin resistant bacterial pathogens belonging to genera *Staphylococcus, Bacillus,* and *Clostridium*.

3.6.1 Mode of Action of Antibiotics

The mode of action of antibiotics varies as they damage pathogens in several ways (Flowchart 3.1). Some of the important actions of therapeutic drugs in microbial pathogens are as follows.

Cell wall synthesis Protein synthesis Nucleic acid synthesis Cell membrane disruption Metabolic pathways blockage

1. Inhibition of Cell Wall Synthesis

The most selective therapeutic antibiotics are those that interfere with the synthesis of bacterial cell walls. These drugs posses a high therapeutic index because bacterial

TN_GOVT_XII_Micro_Biology_CH03.indd 37 15-04-2019 16:35:05